

SSN College of Engineering – Kalavakkam

Department of Computer Science and Engineering

ICS1513 – Computer Networks Laboratory

Experiment 3: Student Record Retrieval Using UDP Socket

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Objective

To implement a connectionless client-server application using UDP datagram sockets in C where the client queries a student record by transmitting a roll number datagram to the server.

Server Algorithm

1. Create a UDP socket using `socket(AF_INET, SOCK_DGRAM, 0)`.
2. Configure the server address structure (`AF_INET, INADDR_ANY, Port 9001`).
3. Bind the UDP socket to the address and port using `bind()`.
4. Enter an infinite loop to receive datagrams using `recvfrom()`, which captures the client's socket address details.
5. Search `students.csv` for the requested roll number using `find_student()`.
6. Format the response buffer with either the matched CSV line or "Student not found\n".
7. Transmit the reply datagram back to the client using `sendto()` with the stored client address details.
8. Repeat the loop to process subsequent client queries.

Client Algorithm

1. Create a UDP datagram socket using `socket(AF_INET, SOCK_DGRAM, 0)`.
2. Set up the target server address structure with `AF_INET, port 9001, and IP address "127.0.0.1"`.
3. Prompt user for roll number, read input via `fgets()`, and trim the trailing newline character.
4. Send the roll number datagram to the server using `sendto()`.
5. Wait for and receive the server's reply datagram using `recvfrom()`.
6. Null-terminate the buffer and display the response.
7. Close the UDP socket using `close()`.

Server Code (udp_student_server.c)

```
#include <stdio.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>

// Search students.csv for the given roll number.
// If found, fill 'result' with the matching line and return 1.
// Otherwise return 0.
int find_student(char *roll, char *result) {
    FILE *fp = fopen("students.csv", "r");
    char line[256];

    if (!fp) return 0;

    fgets(line, sizeof(line), fp); // skip header line

    while (fgets(line, sizeof(line), fp)) {
        char line_copy[256];
        strcpy(line_copy, line);

        char *token = strtok(line_copy, ",");
        if (token != NULL && strcmp(token, roll) == 0) {
            strcpy(result, line);
            fclose(fp);
            return 1;
        }
    }

    fclose(fp);
    return 0;
}

int main() {
    int sock;
    struct sockaddr_in addr, client_addr;
    socklen_t len = sizeof(client_addr);
    char roll[64], result[256], reply[300];

    sock = socket(AF_INET, SOCK_DGRAM, 0);
```

```

addr.sin_family = AF_INET;
addr.sin_addr.s_addr = INADDR_ANY;
addr.sin_port = htons(9001);

bind(sock, (struct sockaddr*)&addr, sizeof(addr));

printf("UDP student server listening on port 9001...\n");

while (1) {
    int n = recvfrom(sock, roll, sizeof(roll) - 1, 0,
                    (struct sockaddr*)&client_addr, &len);
    if (n > 0) {
        roll[n] = '\0';
        printf("Client requested roll no: %s\n", roll);

        if (find_student(roll, result)) {
            strcpy(reply, result);
        } else {
            strcpy(reply, "Student not found\n");
        }

        sendto(sock, reply, strlen(reply), 0,
               (struct sockaddr*)&client_addr, len);
    }
}

close(sock);
return 0;
}

```

Client Code (udp_student_client.c)

```

#include <stdio.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>

int main() {
    int sock;
    struct sockaddr_in addr;
    socklen_t len = sizeof(addr);
    char roll[64], buffer[300];

```

```

printf("Enter roll no: ");
fgets(roll, sizeof(roll), stdin);
roll[strcspn(roll, "\n")] = '\0'; // strip newline

sock = socket(AF_INET, SOCK_DGRAM, 0);

addr.sin_family = AF_INET;
addr.sin_port = htons(9001);
inet_pton(AF_INET, "127.0.0.1", &addr.sin_addr);

sendto(sock, roll, strlen(roll), 0, (struct sockaddr*)&addr, len);

int n = recvfrom(sock, buffer, sizeof(buffer) - 1, 0, NULL, NULL);
if (n > 0) {
    buffer[n] = '\0';
    printf("Server response: %s\n", buffer);
}

close(sock);
return 0;
}

```

Code Output (Test Cases)

Server Output Terminal:

```

UDP student server listening on port 9001...
Client requested roll no: 3122247001057
Client requested roll no: 3122247001999

```

Client Output — Test Case 1 (Valid Record Found):

```

Enter roll no: 3122247001057
Server response: 3122247001057,Saravana Senthilkumar,CSE

```

Client Output — Test Case 2 (Record Not Found):

```

Enter roll no: 3122247001999
Server response: Student not found

```

Learning Outcomes

1. Mastered the fundamentals of connectionless datagram communication using `sendto()` and `recvfrom()`.
2. Understood key operational and architectural distinctions between TCP (connection-oriented, stateful) and UDP (connectionless, stateless) socket programming paradigms.